pydemi 0.3.0

INPUT

CHGCAR | Structure (ML)

MgO_225..._CHGCAR.gz
1.1MB

SUMMARY

Formula	Mg4O4
Atoms	8
Volume (Å³)	76.961
Electrons	32.000
Grid	64 × 64 × 64
Grid points	262,144
a, b, c (Å)	4.2536, 4.2536, 4.2536
α, β, γ (°)	90.00, 90.00, 90.00
Density	DFT (CHGCAR)

SETTINGS

Atomic partition

Nearest

HELP

Descriptor reference (PDF)

pydemi 0.3.0 · Shubham Maurya, CMS Lab, IIT Kanpur. Predicted densities: ChargE3Net, Koker et al., npj Comput. Mater. 2024.

Deploy

Mg4O4 MgO_225_DFT_CHGCAR.gz DFT density

Structure | Descriptors | Plots | Properties | Compare

All 55 | Bonding 18 | Structural 19 | Heterogeneity 18

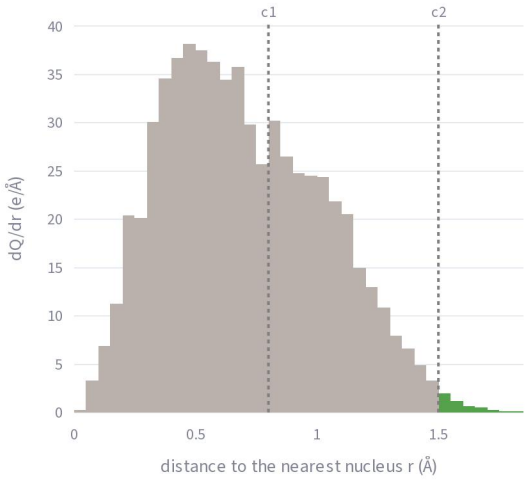
Search name or formula

Export

	descriptor	domain	value	units	fallback
☐	zeta	bonding	0.0242256	dimensionless	☐
☐	m1	bonding	0.703415	Å	☐
☐	m2	bonding	0.600198	Å²	☐
☐	sigma_r2	bonding	0.105405	Å²	☐
☐	f_core	bonding	0.626779	dimensionless	☐
☐	f_bond	bonding	0.366118	dimensionless	☐
☑	f_int	bonding	0.00710331	dimensionless	☐
☐	lnf	bonding	0.0352173	dimensionless	☐
☐	lnf_charge_weighted	bonding	0.364028	dimensionless	☐
☐	lap_concentration_valence	bonding	0	dimensionless	☐
☐	ellip_bond_bounded_avg	bonding	0.247065	dimensionless	☐
☐	f_NCI	bonding	0.164261	dimensionless	☐
☐	NCI_attractive	bonding	0.384673	dimensionless	☐
☐	sign_lambda2_rho_mean	bonding	0.00111573	e/bohr³	☐
☐	V_spread	bonding	60.1729	eV	☐

55 of 55 shown

f_int = 0.007103



Descriptor	f_int
Domain	bonding
Value	0.00710331
Units	dimensionless

FORMULA

$$f_{\text{int}} = \frac{\sum_{k:\, r_k > c_2} \rho_k}{\sum_k \rho_k}$$

Shells around the nearest nucleus: core $r_k \le c_1$, bond $c_1 < r_k \le c_2$,